

2.1 It's always good to mention the definition of $T^{-1}(A)$, $A \subset \Omega_2$, $T^{-1}(A) = \{x \in \Omega_1 : T(x) \in A\}$.

$$\overline{T^{-1}(\bigcup_{\alpha \in I} A_\alpha)} = \bigcup_{\alpha \in I} \overline{T^{-1}(A_\alpha)}$$

$$\begin{aligned} \forall \alpha \in I, A_\alpha \subset \bigcup_{\alpha \in I} A_\alpha &\Rightarrow T^{-1}(A_\alpha) \subset T^{-1}(\bigcup_{\alpha \in I} A_\alpha), \forall \alpha \in I \Rightarrow T^{-1}(\bigcup_{\alpha \in I} A_\alpha) \supset \bigcup_{\alpha \in I} T^{-1}(A_\alpha). \\ \forall x \in T^{-1}(\bigcup_{\alpha \in I} A_\alpha) &\Rightarrow T(x) \in \bigcup_{\alpha \in I} A_\alpha \Rightarrow \exists \alpha_0 \in I, T(x) \in A_{\alpha_0} \Rightarrow x \in T^{-1}(A_{\alpha_0}) \subset \\ &\bigcup_{\alpha \in I} T^{-1}(A_\alpha) \Rightarrow T^{-1}(\bigcup_{\alpha \in I} A_\alpha) \subset \bigcup_{\alpha \in I} T^{-1}(A_\alpha). \end{aligned}$$

$$\overline{T^{-1}(\bigcap_{\alpha \in I} A_\alpha)} = \bigcap_{\alpha \in I} \overline{T^{-1}(A_\alpha)}$$

$$x \in T^{-1}(\bigcap_{\alpha \in I} A_\alpha) \Leftrightarrow T(x) \in \bigcap_{\alpha \in I} A_\alpha \Leftrightarrow T(x) \in A_\alpha, \forall \alpha \in I \Leftrightarrow x \in T^{-1}(A_\alpha), \forall \alpha \in I \Leftrightarrow x \in \bigcap_{\alpha \in I} T^{-1}(A_\alpha).$$

$$\overline{(T^{-1}(A))^c} = T^{-1}(A^c).$$

$$x \in (T^{-1}(A))^c \Leftrightarrow T(x) \notin A \Leftrightarrow T(x) \in A^c \Leftrightarrow x \in T^{-1}(A^c).$$

2.3 Let $\psi(x, y) = \frac{x}{y} \cdot I\{y \neq 0\}$, $\forall a \in \mathbb{R}$,

$$\psi^{-1}((-\infty, a]) = \{(x, y) : x - ay \leq 0, y > 0\} \cup \{(x, y) : x - ay \geq 0, y < 0\} \cup \{(x, y) : 0 \leq a, y = 0\}.$$

That says $\psi^{-1}((-\infty, a]) \in \mathcal{B}(\mathbb{R}^2)$, $\forall a \in \mathbb{R}$, thus $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}$ is $\langle \mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}) \rangle$ -measurable. By propositions in the text book, $\zeta(\omega) \equiv (f(\omega), g(\omega))$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ -measurable. Therefore, $h = \zeta \circ \psi$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

2.6 (a) $A = \{\omega \in \Omega : X_2^2(\omega) - 4X_1(\omega)X_3(\omega) > 0\} \cap \{\omega \in \Omega : X_1(\omega) \neq 0\} \in \mathcal{F}$, because $h(x, y, z) \equiv x^2 - yz$ is continuous, therefore $\langle \mathcal{B}(\mathbb{R}^3), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(b) $\forall \omega \in A$, say

$$T_1(\omega) = \frac{-X_2 + \sqrt{X_2^2 - 4X_1X_3}}{2X_1}, \quad T_2(\omega) = \frac{-X_2 - \sqrt{X_2^2 - 4X_1X_3}}{2X_1}$$

$$\begin{aligned} f_1(\omega) &= T_1(\omega) \cdot I\{\omega : \omega \in A\} \\ &= \frac{(-X_2 + \sqrt{X_2^2 - 4X_1X_3}) \cdot I\{\omega : X_2^2 - 4X_1X_3 > 0\}}{2X_1} \cdot I\{\omega : X_1 \neq 0\} \end{aligned}$$

Clearly, $h(x, y) = (x + \sqrt{y}) \cdot I\{y > 0\} = x \cdot I\{y > 0\} + \sqrt{y} \cdot I\{y > 0\}$ is $\langle \mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}) \rangle$ -measurable by **Corollary 2.1.4**, which means that $g_1(\omega) \equiv (-X_2 + \sqrt{X_2^2 - 4X_1X_3}) \cdot I\{\omega : X_2^2 - 4X_1X_3 > 0\}$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Let $g_2(\omega) = 2X_1(\omega)$, by **Problem 2.3**, $f_1(\omega) = \frac{g_1(\omega)}{g_2(\omega)} \cdot I\{\omega : g_2(\omega) \neq 0\}$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Similarly, $f_2(\omega)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Thus $(f_1(\omega), f_2(\omega))$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ -measurable.

- A. A direct application of **Problem 2.1** will let us see that $\sigma\langle X \rangle \equiv \{X^{-1}(B) : B \in \mathcal{F}_2\}$ is a σ -algebra. For any σ -algebra $\mathcal{G}$ on Ω_2 such that X is $\langle \mathcal{G}, \mathcal{F}_2 \rangle$ -measurable, by definition of measurable function, $\{X^{-1}(B) : B \in \mathcal{F}_2\} \subset \mathcal{G}$, i.e. $\sigma\langle X \rangle \subset \mathcal{G}$.
- B. Take any set $A \notin \mathcal{B}(\mathbb{R})$, define $f(x) = I\{x : x \in A\}$, then f is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable. However, it's actually not trivial to find such an A . One specific example of [Non-Borel set](#) is due to [Lusin](#). For this problem, we can just find any *non-Lebesgue measurable set*, i.e. $A \notin \mathcal{M}_m$, which is actually mentioned in our textbook P. 27. Because $\mathcal{B}(\mathbb{R}) \subset \mathcal{M}_m$, so $A \notin \mathcal{B}(\mathbb{R})$. The [Vitali set](#) is a famous example of non-measurable set.